Graph Theory & Combinatorics

What is Combinatorics?

Feng SHI

石峰

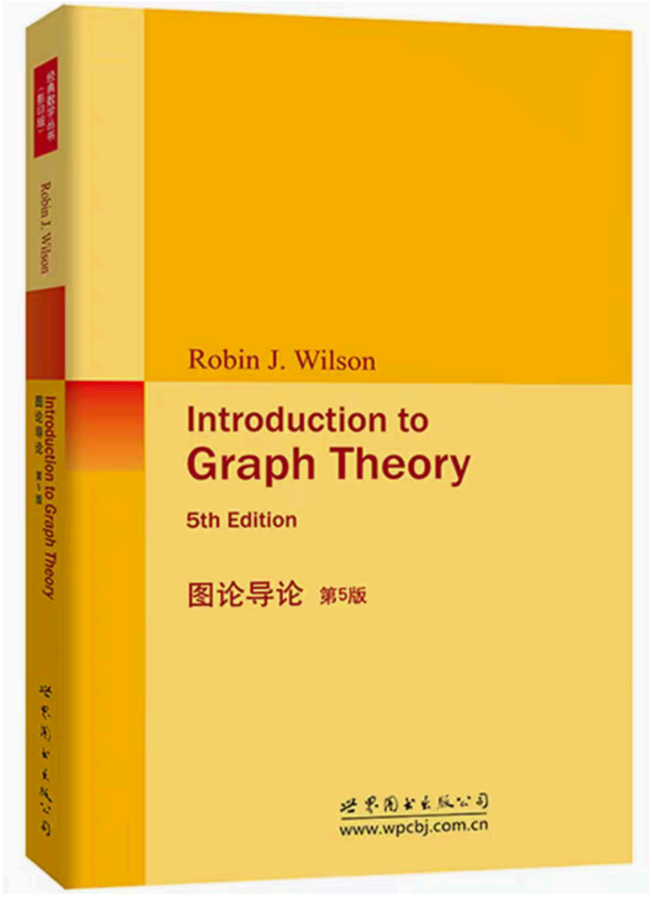
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School of Computer Science and Engineering

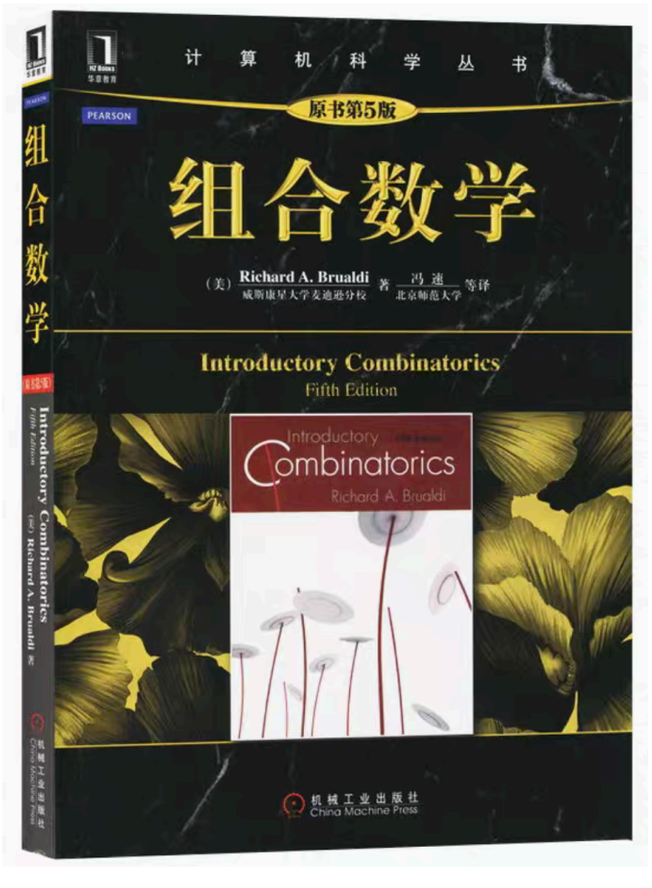
Central South University

课程分数

- 期末考试，60%（闭卷）
- 平时成绩，20%（2-3次课后作业）
- 课堂表现，20%



图论导论 第5版 Introduction to Graph Theory
R.J.威尔逊著



组合数学 原第5版 Richard A Brualdi
计算机科学丛书 黑皮书

THE MAP OF MATHEMATICS

FOUNDATIONS

FUNDAMENTAL RULES

MATHEMATICAL LOGIC

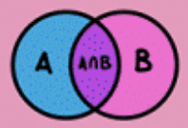
$p \Rightarrow q$

CONSISTENT SET OF AXIOMS?

GÖDEL INCOMPLETENESS THEOREMS

NOPE

SET THEORY





CATEGORY THEORY

THEORY OF COMPUTATION



P≠NP?

COMPLEXITY THEORY

CARDINAL NUMBERS

$\aleph_0$ ALEPH NULL

OCTONION

$\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

QUATERNION

$a+bi+cj+dk$

PI

π

EXPONENTIAL

e

COMPLEX NUMBERS

$3, i, 4+3i, -4i$

PRIME NUMBERS

3, 11, 47, 907

INFINITY

∞



NUMBER THEORY

REAL NUMBERS

$-4\pi, \sqrt{2}, e$

RATIONAL NUMBERS

$-7, \frac{1}{2}, 2.32$

INTEGERS

...-2, -1, 0, 1, 2...

NATURAL NUMBERS

1, 2, 3, 4, 5...

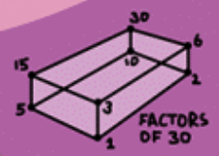
ARITHMETIC $+ - \times \div$

COMBINATORICS

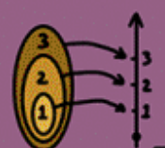
PARTITION THEORY



TREE



ORDER THEORY



MEASURE THEORY

GROUP THEORY



PERMUTATION GROUP

GRAPH THEORY



LINEAR ALGEBRA

$\begin{pmatrix} & \\ & \end{pmatrix} \cdot \begin{pmatrix} & \\ & \end{pmatrix} = \begin{pmatrix} & \\ & \end{pmatrix}$

MATRICES $\begin{pmatrix} 6 & 7 \\ -3 & 2i \end{pmatrix}$

ALGEBRA

$x^2-4x-8=5x+28$

$x^2-9x-36=0$

$(x+3)(x-12)=0$

EQUATION

$y=mx+c$

VECTORS $\vec{x}$

NUMBER SYSTEMS

PURE MATHEMATICS

STRUCTURES

SPACES

CHANGES

ORIGINS

PERSIA c.820

ALGEBRA

INDIA c.628

FIRST ZERO 0

c.1730 MATHEMATICAL NOTATION

$e^{i\pi}=-1$

COUNTING

NEGATIVE NUMBERS

-8 ||| ||

CHINA 200 BCE

50,000 BCE

EGYPT FIRST EQUATION 3000 BCE

GREECE 600-300 BCE



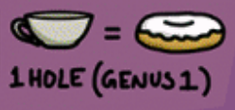


MÖBIUS STRIP

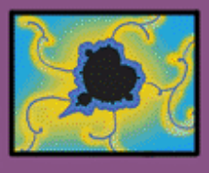


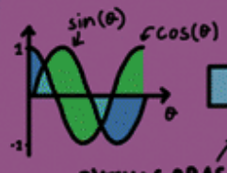
TOPOLOGY

DIFFERENTIAL GEOMETRY

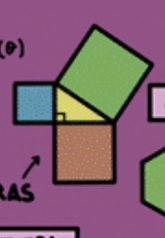


MANDELBROT SET





PYTHAGORAS

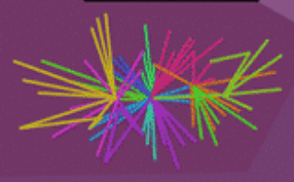


GEOMETRY

TRIGONOMETRY

COMPLEX ANALYSIS

FRACTAL GEOMETRY





BUTTERFLY EFFECT

CHAOS THEORY

DYNAMICAL SYSTEMS

FLUID FLOW



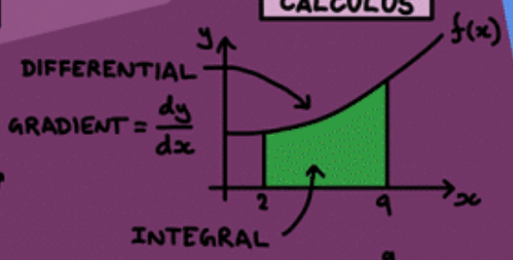


ECOSYSTEMS



VECTOR CALCULUS

CALCULUS



DIFFERENTIAL EQUATIONS

$\text{AREA} = \int_2^9 f(x)\,dx$

APPLIED MATHEMATICS

CRYPTOGRAPHY



COMPUTER SCIENCE

MACHINE LEARNING




```
while awake:
    do_science()
    if self.tired():
        awake = False
self.repair_brain()
```



OPTIMIZATION

PROBABILITY



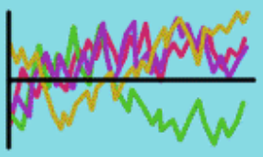
BAYES' RULE

$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

STATISTICS



MATHEMATICAL FINANCE



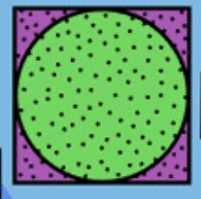
ECONOMICS

GAME THEORY

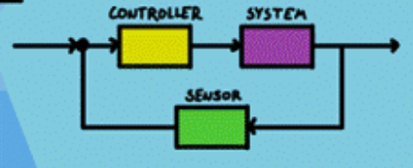




NUMERICAL ANALYSIS

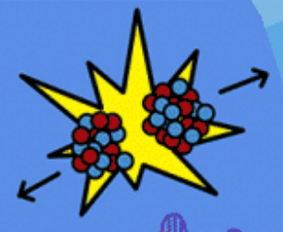


ENGINEERING



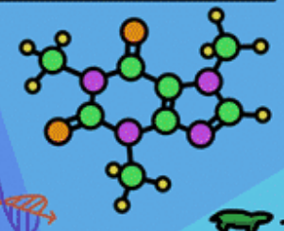
CONTROL THEORY

MATHEMATICAL PHYSICS



THEORETICAL PHYSICS

MATHEMATICAL CHEMISTRY



BIOMATHEMATICS



BY DOMINIC WALLIMAN © 2017

YOUTUBE: THE MAP OF MATHEMATICS

0. Combinatorics

- Combinatorics (组合数学) is a young field of mathematics, starting to be an independent branch only in the 20th century. However, combinatorial methods and problems have been around ever since.

- Many combinatorial problems look entertaining or aesthetically pleasing and indeed one can say that roots of combinatorics lie in mathematical recreations and games.

- Nonetheless, this field has grown to be of great importance in today's world, not only because of its use for other fields like
  - ✓physical sciences
  - ✓social sciences
  - ✓biological sciences
  - ✓information theory
  - ✓computer science

0. Combinatorics

Combinatorics is concerned with:

- Arrangements of elements in a set into patterns satisfying specific rules, generally referred to as discrete structures. Here “discrete” (as opposed to continuous) typically also means finite, although we will consider some infinite structures as well.

- The existence, enumeration, analysis and optimization of discrete structures.

- Interconnections, generalizations- and specialization-relations between several discrete structures.

Existence.

We want to arrange elements in a set into patterns satisfying certain rules.

- Is this possible?
- Under which conditions is it possible?
- What are necessary, what sufficient conditions?
- How do we find such an arrangement?

Enumeration.

Assume certain arrangements are possible.

- How many such arrangements exist?
- Can we say “there are at least this many”, “at most this many” or “exactly this many”?
- How do we generate all arrangements efficiently?

0. Combinatorics

Classification.

Assume there are many arrangements.

- Do some of these arrangements differ from others in a particular way?
- Is there a natural partition of all arrangements into specific classes?

Meta-Structure:

- Do the arrangements even carry a natural underlying structure, e.g., some ordering?
- When are two arrangements closer to each other or more similar than some other pair of arrangements?
- Are different classes of arrangements in a particular relation?

Optimization.

Assume some arrangements differ from others according to some measurement.

- Can we find or characterize the arrangements with maximum or minimum measure, i.e. the “best” or “worst” arrangements?

Interconnections.

Assume a discrete structure has some properties (number of arrangements, . . .) that match with another discrete structure.

- Can we specify a concrete connection between these structures?
- If this other structure is well-known, can we draw conclusions about our structure at hand?

We will give some life to this abstract list of tasks in the context of the following example.

Example (Dimer Problem).

Consider a generalized chessboard of size $m \times n$ (m rows and n columns). We want to cover it perfectly with dominoes of size 2×1 or with generalized dominoes – called polyominoes – of size $k \times 1$. That means we want to put dominoes (or polyominoes) horizontally or vertically onto the board such that every square of the board is covered and no two dominoes (or polyominoes) overlap. A perfect covering is also called tiling. Consider Figure 1 for an example.

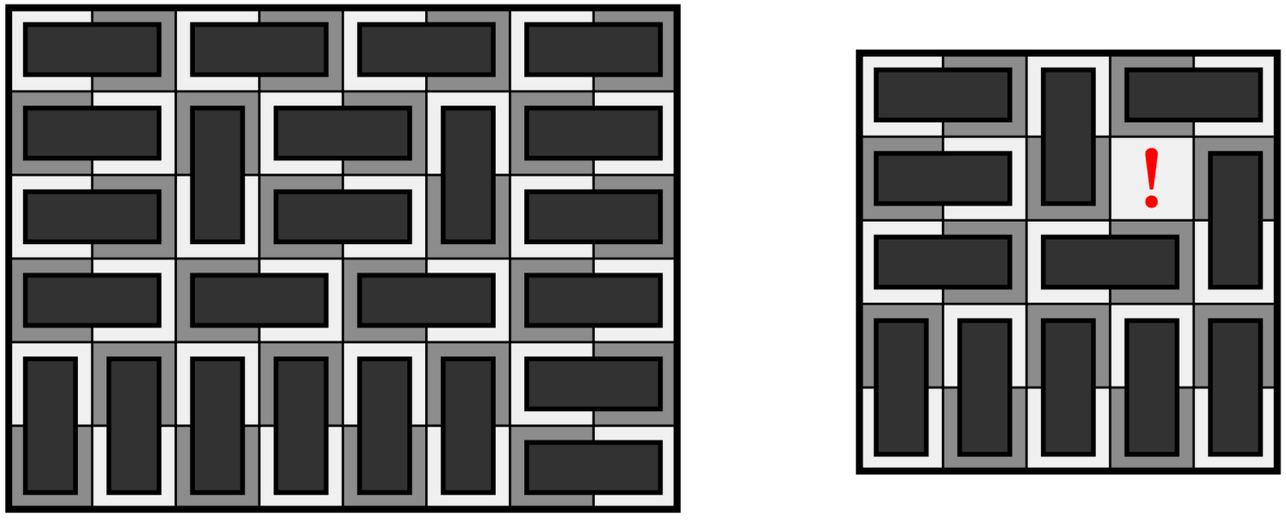


Figure 1: The 6 × 8 board can be tiled with 24 dominoes. The 5 × 5 board cannot be tiled with dominoes.

Existence

If you look at Figure 1, you may notice that whenever m and n are both odd (in the Figure they were both 5), then the board has an odd number of squares and a tiling with dominoes is not possible. If, on the other hand, m is even or n is even, a tiling can easily be found. We will generalize this observation for polyominoes:

Claim.

An $m \times n$ board can be tiled with polyominoes of size $1 \times k$ if and only if k divides m or n.

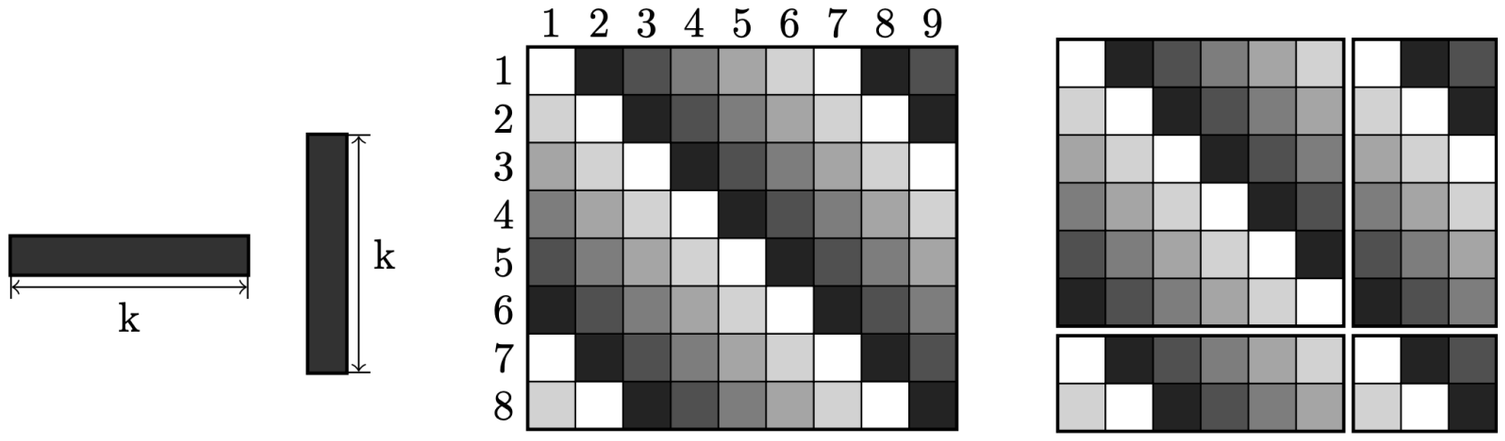


Figure 2: Our polyominoes have size $k \times 1$. We use k colors (1 = white, k = black) to color the $m \times n$ board (here: $k = 6$, $m = 8$, $n = 9$). Cutting the board at coordinates that are multiples of k divides the board into several chunks. All chunks have the same number of squares of each color, except for the bottom right chunk where there are more squares of color 1 (here: white) than of color 2 (here: light gray).

Claim.
An $m \times n$ board can be tiled with polyominoes of size $1 \times k$ if and only if k divides m or n.

Proof. "⇐" If k divides m, it is easy to construct a tiling: Just cover every column with m/k vertical polyominoes. Similarly, if k divides n, cover every row using n/k horizontal polyominoes.

"⇒" Assume k divides neither m nor n (but note that k could still divide the product $m \cdot n$). We need to show that no tiling is possible. We write $m = s_1 k + r_1$, $n = s_2 k + r_2$ for appropriate $s_1, s_2, r_1, r_2 \in N$ and $0 < r_1, r_2 < k$. Without loss of generality, assume $r_1 \leq r_2$ (the argument is similar if $r_2 < r_1$). Consider the coloring of the $m \times n$ board with k colors as shown in Figure 2.

Formally, the color of the square (i, j) is defined to be $((i - j) \mod k) + 1$. Any polyomino of size $k \times 1$ that is placed on the board will cover exactly one square of each color. However, there are more squares of color 1 than of color 2, which shows that no tiling with $k \times 1$ dominoes is possible. Indeed, for the number of squares colored with 1 and 2 we have:

squares colored with 1 = $ks_1s_2 + s_1r_2 + s_2r_1 + r_2 - 1$

squares colored with 2 = $ks_1s_2 + s_1r_2 + s_2r_1 + r_2 - 2$ □

Claim (Mutilated Chessboard).

The $n \times n$ board with bottom-left and top-right square removed (see Figure 3) cannot be tiled with (regular) dominoes.

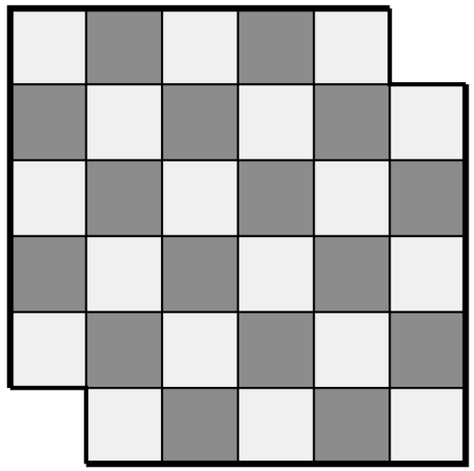


Figure 3: A “mutilated” 6×6 board. The missing corners have the same color.

Proof. If n is odd, then the total number of squares is odd and clearly no tiling can exist. If n is even, consider the usual chessboard-coloring: In it, the missing squares are of the same color, say black. Since there was an equal number of black and white squares in the non-mutilated board, there are now two more white squares than black squares. Since dominoes always cover exactly one black and one white square, no tiling can exist. □

Enumeration

A general formula to determine the number of ways an $m \times n$ board can be tiled with dominoes is known. The special case of an 8×8 board is already non-trivial:

Theorem (Fischer 1961).
There are $2^4 \cdot 17^2 \cdot 53^2 = 12{,}988{,}816$ ways to tile the 8×8 board with dominoes.

Classification

Consider tilings of the 4 × 4 board with dominoes. For some of these tilings there is a vertical line through the board that does not cut through any domino. Call such a line a vertical cut. In the same way we define horizontal cuts.

As it turns out, for every tiling of the 4 × 4 board at least one cut exists, possibly several (try this for yourself!).

Hence the set T of all tilings can be partitioned into

$$T_1 = \{T \mid T \text{ allows a horizontal cut but no vertical cut}\},$$
$$T_2 = \{T \mid T \text{ allows a vertical cut but no horizontal cut}\},$$
$$T_3 = \{T \mid T \text{ allows both a horizontal and a vertical cut}\}.$$

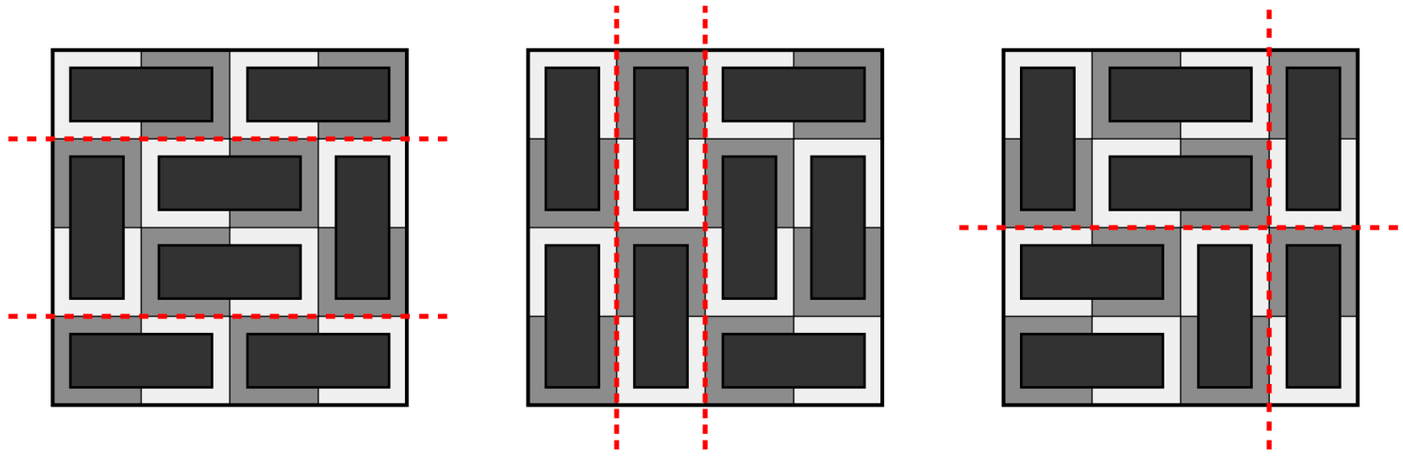


Figure 4: Some tilings have horizontal cuts, some have vertical cuts and some have both.

Classification

As another example consider the board B consisting of two 4×4 boards joint by two extra squares as shown in Figure 5.

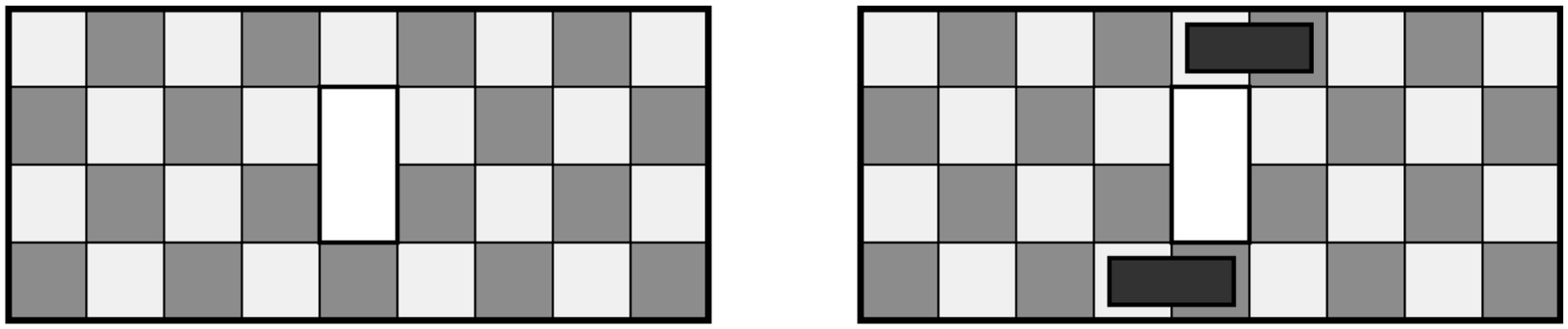


Figure 5: The board B consisting of two 4×4 boards and two extra squares connecting them as shown. The partial covering on the right cannot be extended to a tiling.

In a tiling of B it is impossible for the extra squares to be matched "to different sides": If we place the two dominoes as shown on the right in Figure 5, we are left with two disconnected parts of size 15. And since 15 is odd, this cannot work out. Hence the set T of all tilings of B can be partitioned into

$$T_1 = \{T \mid T \text{ matches both extra squares to the left}\},$$

$$T_2 = \{T \mid T \text{ matches both extra squares to the right}\}.$$

Meta Structure

We say two tilings are adjacent, if one can be transformed into the other by taking two dominoes lying like this 〓 and turn them by 90 degrees so they are lying like this 𝍐 (or vice versa). Call this a turn operation. If we draw all tilings of the 4×4 board and then draw a line between tilings that are adjacent, we get the picture on the left of Figure 6.

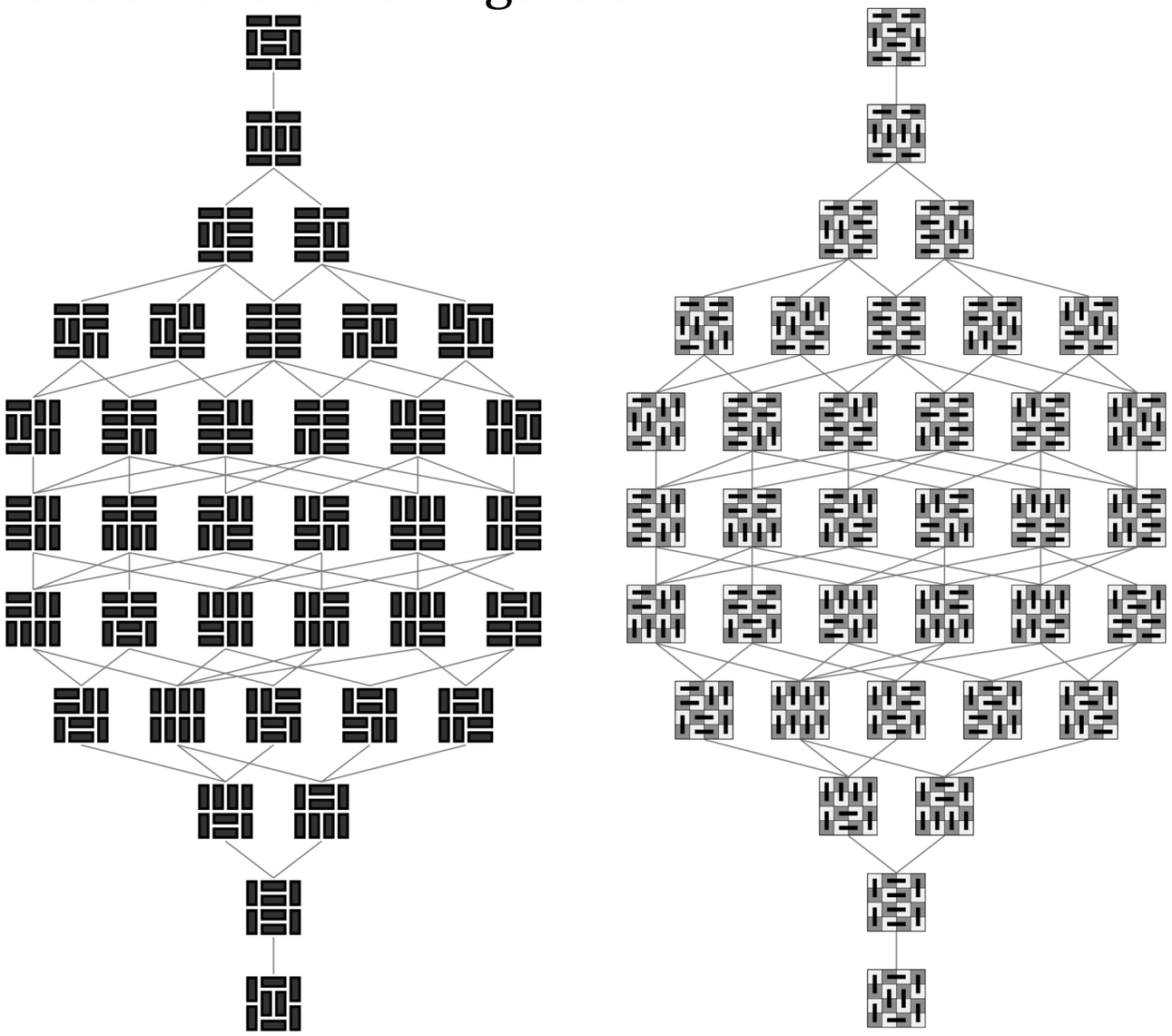


Figure 6: On the left: The set of all tilings of the 4×4 board. Two tilings are connected by an edge if they can be transformed into one another by a single flip. On the right: The same picture, but with the chessboard still drawn beneath the tilings, so you can check that upward edges correspond to flips (as defined in the text).

Meta Structure

- With this in mind, we can speak about the distance of two tilings, the number of turn operations required to transform one into the other.
- We could also look for a pair of tilings with maximum distance (this would be an optimization problem).
- We can even discover a deeper structure, but for this we need to identify different types of turn operations. We can turn 〓 into ▮▮, call this an up-turn, or ▮▮ into 〓, call this a side-turn.
- The turn can happen on the background ▩ or the background ▩ (white squares form a falling or rising diagonal).
- We call an operation a flip if it is an up-turn on ▩ or a side-turn on ▩. Call it a flop otherwise.
- Quite surprisingly, walking upward in the graph in Figure 6 always corresponds to flips and walking downwards corresponds to flops.
- This means that there is a natural partial ordering on the tilings: We can say a tiling B is greater than a tiling A if we can get from A to B by a sequence of flips.
- As it turns out, the order this gives has special properties. It is a so-called distributive lattice, in particular, there is a greatest tiling.

Optimization

Different tilings have a different set of decreasing free paths. Such a path (see Figure 7) proceeds monotonously from the top left corner of the board along the borders of squares to the bottom right, and does not “cut through” any domino.

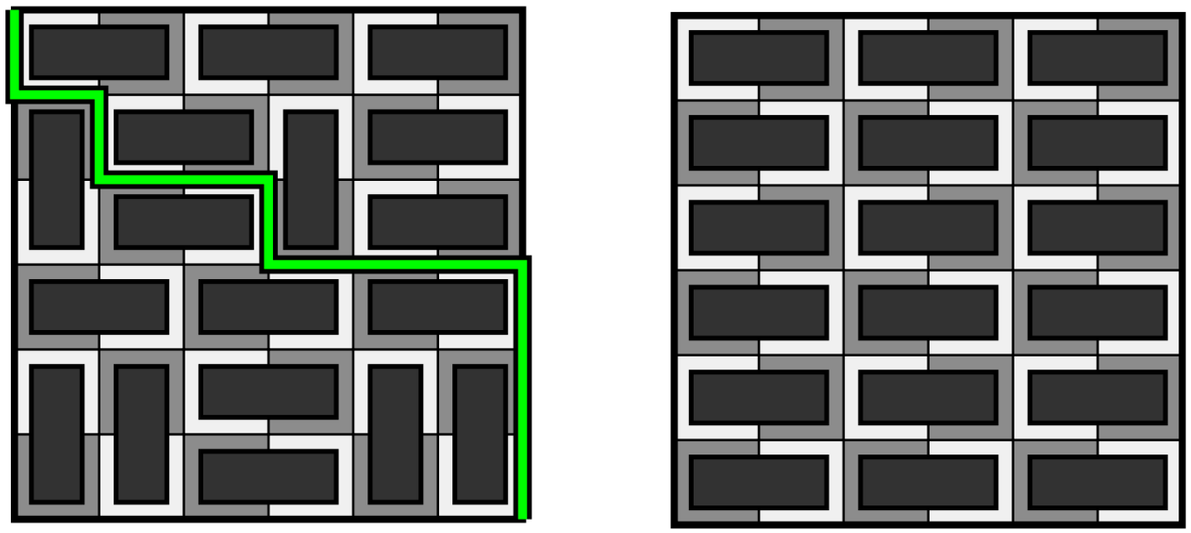


Figure 7: The green path does not cut through dominoes. We believe that the boring tiling on the right admits the maximum number of such paths, but this has not been proved for large boards.

Some tilings admit more such paths than others. It is conjectured that in an $m \times n$ board the maximum number is always attained by one of the two tilings where all dominoes are oriented the same way (see right of Figure 7). But no proof has been found yet.

Thanks for your attention!

Any question?